Attention-Driven Dual Feature Guidance for Hyperspectral Super-Resolution

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***Abstract*— Benefiting from the high spectral resolution, hyperspectral image (HSI) owns the property of discriminating material. However, the spatial resolution of HSIs is limited by the hardware and, thus, makes HSI super-resolution (SR) a necessary and hot topic. Existed HSI SR methods rarely consider the high-frequency edge information, resulting in unsatisfactory quality of spatial reconstruction. To address the problem mentioned above, we propose a novel method for HSI SR named attention-driven dual feature guidance net (AD-DFGNet), which makes full use of the spatial–spectral information. Specifically, AD-DFGNet mainly consists of three modules, which are shallow feature extraction, DFG blocks, and upsampling fusion module. First, the three adjacent bands of an HSI cube are selected sequentially as one of the inputs to the proposed AD-DFGNet. Their feature dimension is upgraded through shallow feature extraction. Second, middle band of three adjacent bands is used as the other input to DFG block, and it makes the network focus on spatial feature extraction by DFG, which includes feature aggregation guidance (FAG) and gradient–texture attention (GTA) guidance. Then, the output features of DFG blocks are upsampled and fused, in turn, to obtain the single-band SR result. Finally, the SR results of the whole HSI are obtained from the sequential concatenation of each single band. In addition, the proposed method reduces the size of the generated model and makes it possible to apply on different datasets. The efficiency of the AD-DFGNet is validated on three publicly available hyperspectral datasets and yields the state-of-the-art performance.**

***Index Terms*— Dual feature guidance (DFG), gradient–texture attention (GTA), hyperspectral image (HSI), super-resolution (SR).**

I. Introduction

A HYPERSPECTRAL image (HSI) is a 3-D data cube, which consists of tens or even hundreds of bands [1]. HSIs can be reflected in the spatial dimension as the spatial arrangement of objects in each band, while, in the spectral dimension, each pixel can be described as an approximately continuous spectral curve with the ability to discriminate different substances [2]. HSIs are widely used in numerous fields, such as for military rescue [3], environmental monitoring [4], and so on [5], [6], [7]. However, limited by the imagery sensors, HSI is characterized by low spatial resolution, resulting in inaccurate interpretation. Thus, super-resolution (SR) is used to improve spatial resolution and provide higher quality images for other tasks, such as classification [8], detection [9], [10], and segmentation [11].

HSI SR has been increasingly explored as a popular topic in recent year. Many scholars have proposed different approaches to solve the low spatial resolution of HSIs, and these approaches can be classified into two categories. The first category is fusion-based method, which combines the advantages of multisource images and reconstructs the desired spatial–spectral feature of HSIs. The second category is single-to-single method, which generates SR results by learning the spatial–spectral mapping relationship between low-resolution (LR) HSI and high-resolution (HR) HSI.

In general, multispectral image (MSI), panchromatic (PAN) image, and RGB image are the HR inputs of the fusion-based methods, because these images mentioned above have rich spatial features and provide sufficient spatial prior information in the SR process. Zhao et al. [12] used regularization of spatial and spectral nonlocal similarities to fuse the sparse regulation of HSI-PAN image. Dong et al. [13] proposed an efficient nonnegative dictionary learning algorithm, which used the block-coordinate descent optimization technique with LR-HSI and HR-RGB as inputs. Lanaras et al. [14] combined LR-HSI with conventional HR-MSI to obtain SR result by jointly unmixing the two images into pure reflectance spectral of the observed materials, along with the associated mixing coefficients. To address the drawback that matrixization tends to lose structural information, Kanatsoulis et al. [15], Dian et al. [16], Dian and Li [17], and Xu et al. [18] utilized the tensor decomposition instead of the matrix sparse representation. In addition, deep learning methods can obtain excellent results on pansharpening. Zheng et al. [19] used deep hyperspectral prior and dual-attention residual network to deal with LR-HSI and fusion images, respectively. Bandara et al. [20] observed that the quality of upsampled HSIs can be further improved by introducing an additional spatial-domain constraint to the conventional spectral-domain energy function. Recently, Bandara and Patel [21] improved both spatial and spectral quality measures of the pansharpened HSI by learning cross-feature space dependencies and long-range details of PAN and LR-HSI. The fusion-based method incorporates the different characteristics of multisource images, and it could effectively improve the spatial resolution of LR-HSIs. However, the fusion-based method is strict in requiring fully

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registered input images, which limits the practicability of the fusion-based SR methods.

Due to the facility of single HSI for SR, single-to-single strategy has become a hot research topic in recent years. Convolutional neural network (CNN) has the advantage of extracting deep features from images and in end-to-end self-learning. In the field of supervised learning, it can effectively learn the mapping relationship between LR-HSI and HR-HSI. Since the use of super-resolution convolutional neural network (SRCNN) [22] in single-image SR, a large number of CNN-based methods have been inspired to super-resolve the HSI-SR as well. To consider the spectral reconstruction performance of the network, Mei et al. [23] used 3-D convolution to learn the spectral feature mapping relationship between LR-HSI and HR-HSI, which reduced the spectral distortion of the HSI effectively. Hu et al. [24] utilized spectral different convolution neural network (SDCNN) to fully explore the interspectral and spatial correlations and achieved good spatial information enhancement and spectral information retention. Based on the proposed SDCNN, Li et al. [25] combined spatial constraint with the advantages of SDCNN to reconstruct HR-HSI with more excellent spatial–spectral quality. Xie et al. [26] combined the nonnegative matrix decomposition strategy with the SDCNN model and, therefore, achieved a significant role in simultaneously extracting nonnegative spatial–spectral features. Li et al. [27] proposed grouped deep recursive residual network without using auxiliary images, which better described complex nonlinear mapping functions with compact network. Jiang et al. [28] introduced a group convolution and progressive upsampling network with shared network parameters, and it had spatial–spectral blocks, which consists of a spatial residual module and a spectral attention residual module. Furthermore, HSI SR has new applications in remote sensing, such as [29], [30], and [31]. Arun et al. [32], Hu et al. [33], Li et al. [34], Dou et al. [35], Liu et al. [36], and Yang et al. [37] used 3-D convolution in the network structure, which largely maintained the spectral characteristics of HSI. But, we all know that 3-D convolution leads to dramatically larger network models. Li et al. [38], [39] and Wang et al. [40] proposed a mixture of 2-D and 3-D convolution, which was used to maintain the desired network performance while reducing the network parameters. Li et al. [39] explored the relationship between 2-D and 3-D convolution for HSI-SR. Wang et al. [40] utilized the contextual feature information of HSIs along with the 3-D convolution strategy, resulting in the state-of-the-art performance. Furthermore, Hu et al. [41] paid more attention to how efficiently exploit the spectral information among consecutive LR-HSI bands and proposed an intrafusion network to generate high spatial–spectral quality SR HSIs. Dong et al. [42] constructed a devoted subnetwork for approximating the degradation matrix to further optimize the constructed model-guided deep convolution network in an end-to-end manner. Deep learning strategies demonstrate strong potential in a single-to-single method. However, the above single HSI-based SR method used 3-D convolution, which leads to a large number of parameters and is not conducive to model training.

In summary, the existed methods of SR reconstruction suffer from the following problems.

1) The issue of spectral distortion has not been thoroughly considered. Even if the existed 3-D convolution could alleviate the problem mentioned, the interband features entering the network are too huge to consider adequately, leading to severe global and local spectral distortions.
2) The extraction of spatial feature may contradict with the extraction of spectral feature. Most methods use either a serial operations or parallel approach to extract both the spatial and spectral features. However, both extractions will constrain the other one. This constraint greatly hinders the effectiveness of the network in learning another feature.

In order to obtain HR-HSIs with desired spatial quality, we propose a novel HSI-SR method in this article named attention-driven dual feature guidance net (AD-DFGNet) to make a thorough exploration in the space and spectrum. In the proposed AD-DFGNet, an attention mechanism is introduced in the gradient–texture attention (GTA) to enhance the edge information of the reconstructed image. This operation utilizes gradient and texture details to improve the spatial quality of the image. Specifically, GTA is combined with the depth aggregation feature extraction of LR-HSI to achieve DFG for the SR process. Meanwhile, in order to reduce the spectral reconstruction error of HSI, the three adjacent bands are the input of the network, which effectively maintains the local continuity between bands. This scheme has been experimentally shown to be efficient in terms of performance improvement. In summary, the main contributions of the proposed method can be summarized as follows.

1) Two vital branches are applied to the network. The proposed method enables the three adjacent bands and middle band to be used as input to the backbone branch and guide branch, respectively, which takes a full account of spatial information as well as local spectral correlations. Meanwhile, the spectral distortion is alleviated significantly. As far as we know, this is the first time that the strategy is applied to HSI-SR.
2) Attention-driven gradient and texture information are introduced into the reconstruction process, which effectively provides superior constraints on the spatial reconstruction of LR-HSI. Gradient–texture information is employed to guide the reconstruction process and, thus, super-resolves better spatial information.
3) The proposed AD-DFGNet achieves the best performance in terms of objective evaluation metrics. Meanwhile, visual effects are optimal additionally by conducting experiments on three different kinds of publicly available datasets.

The rest of this article are organized as follows. Section II makes a description of the existed HSI-SR works. Section III elaborates on the details of the proposed AD-DFGNet. Experimental setup and discussion about the experimental results have been demonstrated in Section IV. Finally, Section V concludes this article.

II. Related Works

In this section, the works related to the proposed AD-DFGNet are presented, which includes 2-D/3-D-based

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HSI-SR methods, attention-driven HSI-SR methods and gradient-guided HSI-SR methods.

A. 2-D/3-D-Based HSI-SR Methods

In the face of the HSI-SR problem, CNNs can effectively extract the spatial features of the input HSI. However, when considering the spectral information preservation of the reconstructed HSI, the traditional 2-D convolution has no advantage in comparison with other methods, which use correlation between spectra. Hu et al. [24] proposed SDCNN, which takes full account of the spatial–spectral correlation and makes some progress in spatial and spectral reconstruction using 2-D convolution. Obviously, although the 2-D is efficient in spatial feature extraction, it cannot fully exploit the correlation in the spectral domain, especially for the HSIs with large numbers of bands. In this way, the 3-D convolution is involved in the HSI SR process. Mei et al. [23] improved the SRCNN by using 3-D convolution to completely replace 2-D convolution and by analyzing and improving the number of feature layers. It resulted in a new level of retention of SR spectral correlation. Since then, the 3-D convolution has been heavily used in HSI-SR. A series of methods were proposed by Li et al. [38], [39] and Wang et al. [40] to further explore how 3-D convolution could be better used in HSI-SR. In order to reduce the number of parameters while maintaining comparable performance, they used separable 3-D convolution instead of the traditional 3-D convolution. However, the large computational volume of 3-D convolution does not provide an appreciable advantage in terms of time complexity.

B. Attention-Driven HSI-SR Methods

Attention mechanism has been widely used in the field of computer vision after the squeeze-and-excitation networks (SENet) [43] and self-calibrated networks (SCNet) [44] were proposed. This mechanism enables the extraction of high-level semantic attention information through the focus operation, which is achieved through squeeze and excitation. Benefiting from the attention mechanism, generative adversarial network (GAN) [34] used band attention to maintain spectral correlation and better performance in the case of large-upsampling factor. The 3-D attention with GAN [35] expanded the feature dimension, and attention operation is performed on the 3-D feature space to make the features more global. Residual dense network (RDN) attention-driven method utilized the RDN [36] as the backbone, and the attention modules are applied to the spectral information reconstruction process, which also greatly reduced the spectral distortion and well recovered the spatial information. The essence of the application of the attention mechanism in the HSI-SR is to better focus on those regions where features are more pronounced and to better guide the recovery of better spatial and spectral information within these regions.

C. Gradient-Guided HSI-SR Methods

Image SR reconstruction methods often have limited performance at the image edges [45]. The gradient image is computed from an edge extraction function (e.g., Roberts operator, Sobel operator, Prewitt operator, and so on) which reflects the global structural information of the image. Finely reconstructing the gradient edges and guiding the reconstruction process of the original image are effective in enhancing the visual effect of SR. In single-image SR task, Sun et al. [45] pioneered the real scene of extracting gradient information of images for SR using the traditional method, which constructed an objective function based on Kullback-Leibler (KL) scatter by analyzing the gradient magnitude variation of images. Based on [45], [46] and [47] reconstructed the method of describing the image edge information in different approaches, which described the variation of the edge information more precisely. Zhao et al. [48] used a deep learning approach by adding the gradient images extracted from the original HSI to the training process of the network as well to guide the reconstruction of SR and achieved promising results. Hu et al. [49] improved the bootstrapping process of gradient information, so that HSIs were decomposed into two parts, which are gradient and texture. It used edge information and detail information to act together in parallel. In the proposed AD-DFGNet, the single-band HSI with deep feature extraction is divided into a gradient map and a texture map. In order to reconstruct the obvious details of the HSI, the edge information and the texture information are fully utilized in the proposed method.

III. Proposed Method

The overall architecture of the proposed network is shown in Fig. 1. The input of each process is three bands, which contains the current band and two adjacent bands. The output is the generated SR image of the current band. Four main parts of the proposed method are presented: adjacent bands extraction module, single-band SR module, bands concatenation module, and loss function. Detailed descriptions about these four parts are presented in Sections III-A–III-D.

A. Adjacent Bands Extraction Module

The more distant the two bands are, the greater the spectral variability is. In order to alleviate the large amount of data input from the HSI, the proposed AD-DFGNet takes the current band and its neighboring bands as the input, which maintains the spectral correlation of the neighboring bands. The specific division method is listed as follows:

$$I_{\text{in}} = \begin{cases} \text{Concat}[I_i, I_i, I_{i+1}], & i = 1 \\ \text{Concat}[I_{i-1}, I_i, I_i], & i = B \\ \text{Concat}[I_{i-1}, I_i, I_{i+1}], & \text{otherwise} \end{cases} \tag{1}$$

where I is input LR-HSI, I_{in} is three adjacent bands as the input of network, and B represents the band number of the LR-HSI.

Separated bands mitigate the mutual exclusion of uncorrelated spectral information and alleviate the interconstraints with spatial information in end-to-end learning.

B. Single-Band SR Module

In this section, the network framework is described in detail. It is mainly divided into three steps: shallow feature extraction, AD-DFG, and upsampling fusion.

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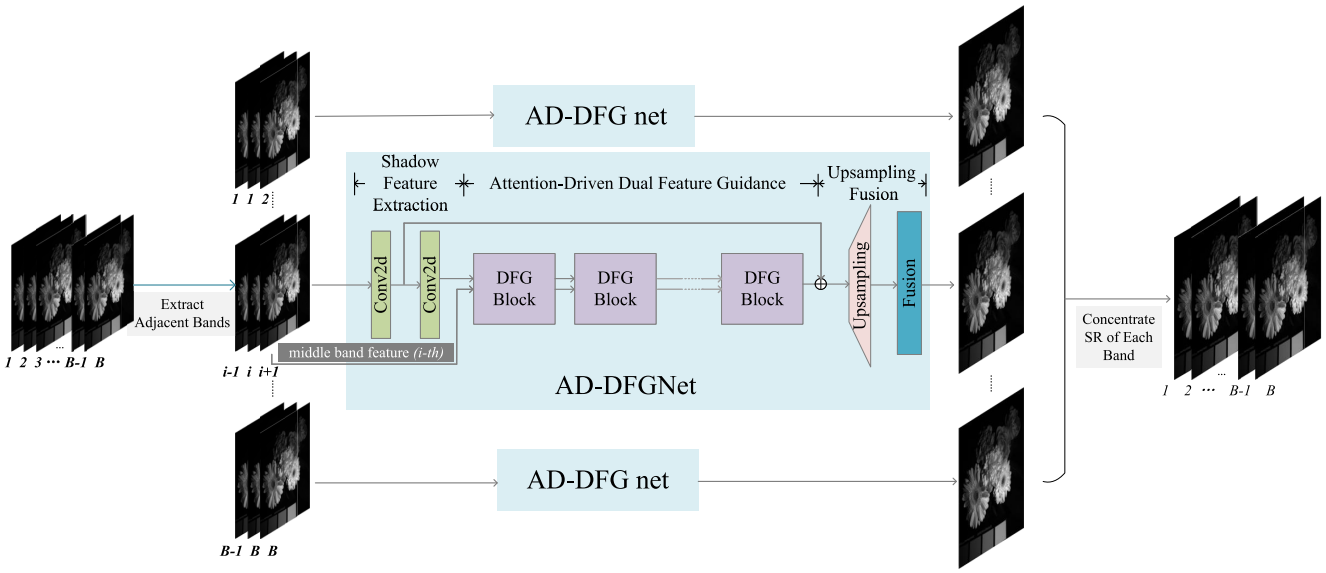


Fig. 1. Framework of the proposed AD-DFGNet.

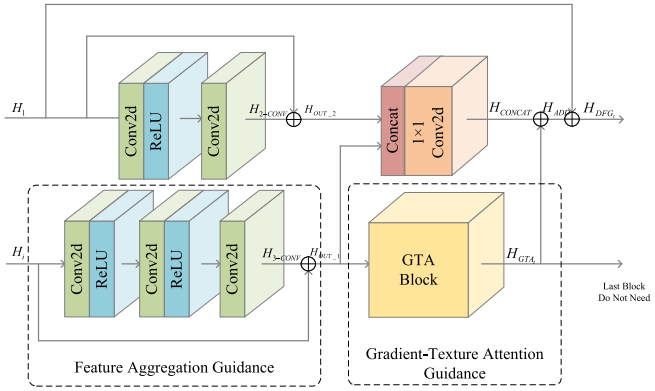


Fig. 2. Architecture of the proposed DFG module.

1) Shallow Feature Extraction: In this step, in order to perform feature extraction in a high-dimensional feature space, a 2-D convolutional layer is first used to up dimension the 3-D input, so that the original features are mapped to higher dimension. Specifically, the original features with three channels are mapped to the higher dimension with 64 channels. This process is denoted as follows:

$$H_0 = \sigma(\text{Conv}(I_{\text{in}})) \tag{2}$$

where Conv() and $\sigma()$ are defined as the convolution layer and activation function ReLU(), respectively. Then, an additional 2-D convolution is used to perform a shallow feature extraction on the upscaled feature cube

$$H_1 = \sigma(\text{Conv}(H_0)) \tag{3}$$

where H_1 is represented as output of the shallow feature extraction module.

2) Attention-Driven DFG: As shown in Fig. 2, each block in the AD-DFG module contains two branches: the upper branch is the reconstruction branch, and the lower branch is the guidance branch. Reconstruction branch maintains the complete reconstruction process of the network, while the guidance branch acts as the main branch for extracting spatial features of the image. The dual guidance branch could inject rich feature information into the main branch. It greatly enhances the effect of feature extraction and is mainly divided into feature aggregation guidance (FAG) and GTA guidance. Meanwhile, the combination of the dual-branch architecture with the adjacent bands extraction strategy improves the concentration of the spatial–spectral feature extraction and mitigates the negative impact of the different features. In order to effectively improve the spatial reconstruction of the current band, a separate band H_i is used as input in guidance branch to avoid the interference of other bands, which can be expressed as follows:

$$H_{\text{OUT_1}} = \text{FAG}(H_i) \tag{4}$$

where FAG() contains three layers of 2-D convolution for spatial feature extraction and a residual concatenation operation, which aims at deeply explorating spatial information of the input ith feature and achieves the first guidance of the reconstruction branch. In the GTA block, to deeply exploit the high-frequency details in SR reconstruction, we use an edge extraction function to generate the gradient and texture of the HSIs, and achieve edge information and overall texture information separately. The generation process can be described as follows:

$$H_{\text{grad}}, H_{\text{tex}} = \ell(H_{\text{OUT_1}}) \tag{5}$$

where H_{grad} and H_{tex} are the edge image and the texture image obtained by edge extraction function $\ell()$, respectively. In the design of the GTA, two convolution layers are first utilized to extract the edge and texture features, which is defined as follows:

$$\begin{aligned} H_{g_\text{CONV}} &= \text{Convs_}G(H_{\text{grad}}) \\ H_{t_\text{CONV}} &= \text{Convs_}T(H_{\text{tex}}) \end{aligned} \tag{6}$$

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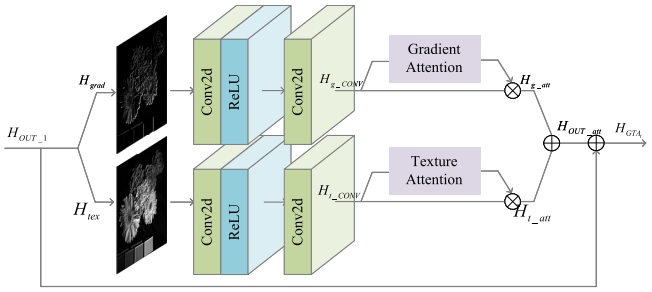


Fig. 3. Architecture of the proposed GTA module.

where Convs_$G()$ and Convs_$T()$ denote the feature extraction for gradient image and texture image, respectively. In addition, the attention mechanism provides a way to focus features on the parts that are more desirable to be processed. Gradient attention (GA) and texture attention (TA) achieve acceptable highlight of edge features and overall enhancement of structure features, which is shown in Fig. 3. The above process can be described as follows:

$$\begin{aligned} H_{g_\text{att}} &= \text{GA}(H_{g_\text{CONV}}) \\ H_{t_\text{att}} &= \text{TA}(H_{t_\text{CONV}})) \end{aligned} \tag{7}$$

where H_{g_att} and H_{t_att} are the outputs after performing GA and TA, respectively. Fig. 4(a) and (b) shows the feature maps of gradient image and texture image through GA and TA, respectively. The fusion of the two attentional components with the residuals of this module forms the output of the GTA block

$$H_{\text{GTA}} = (H_{g_\text{att}} + H_{t_\text{att}}) \oplus H_{\text{OUT_1}} \tag{8}$$

where H_{GTA} is used as the output of the GTA block for a second guidance of the reconstruction branch.

DFG details are described as follows. In the reconstruction branch, we describe the process of dual guidance in detail. First, two convolutional layers to further extract the features of the image input to the DFG block are defined as follows:

$$H_{\text{OUT_2}} = \text{Conv}(\sigma(\text{Conv}(H_1)))\widehat{\oplus}H_1 \tag{9}$$

where $H_{\text{OUT_1}}$ is the output of the two convolution layers and $\widehat{\oplus}$ is denoted as the local residual add operation. Then, the first feature guidance is performed to concatenate $H_{\text{OUT_1}}$ with the output $H_{\text{OUT_2}}$ of the FAG in the band dimension, and then, the features are compressed by a 1×1 convolutional layer to fuse the features of the two parts. Therefore, the reconstruction branch can utilize more spatial information extracted by the FAG, which can be expressed as follows:

$$H_{\text{CONCAT}} = \text{Conv1}(\text{Concat}([H_{\text{OUT_1}}, H_{\text{OUT_2}}])). \tag{10}$$

Then, the GTA block combines the gradient/texture of traditional image processing methods and deep learning extracted features. Furthermore, attention mechanism is introduced to deeply explore the gradient and textural prior information in HSI-SR task. The secondary guidance can be formulated as follows:

$$H_{\text{ADD}} = H_{\text{CONCAT}} + H_{\text{GTA}_t} \tag{11}$$

where H_{ADD} is the output of the secondary guidance, which combines the block global residuals of the reconstructed

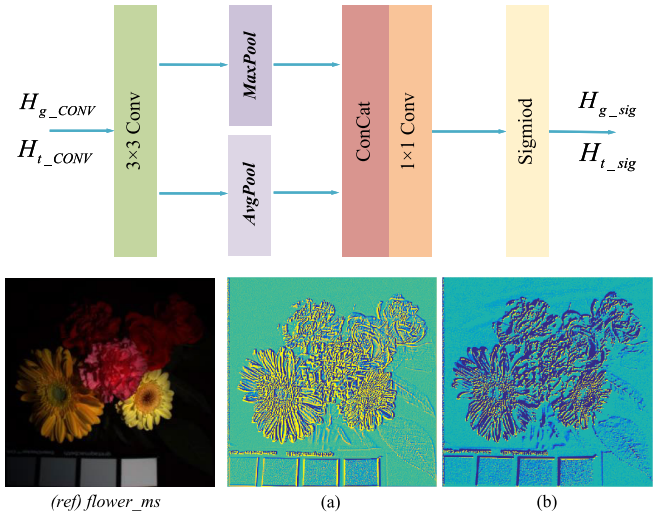


Fig. 4. GA and TA mechanism. MaxPool and AvgPool acquire the local and global information of the input feature layers, respectively, which are then connected by CONCAT to fully fuse the features and finally use Sigmoid to focus on salient information (e.g., edge and texture). Ref: original image. (a) and (b) Feature maps of gradient image and texture image through GA and TA, respectively.

branch to complete the feature fusion of the DFG block and export the final feature H_{DFG_t}

$$H_{\text{DFG}_t} = H_{\text{ADD}} + H_{\text{DFG}_{t-1}}. \tag{12}$$

3) Upsampling Fusion: After extracting features of DFG blocks, the global residual is used to enable the network to be easily trained, which can be defined as follows:

$$H_L = H_{\text{DFG}_T} + H_0 \tag{13}$$

and then use pixelshuffle to upsample the output H_L to the target scale

$$H_{\text{UP}} = \text{up}(H_L). \tag{14}$$

In order to generate an HR-HSI of the current single band, it is necessary to compact the high-dimensional features into the reconstructed single band at last, which can be defined as follows:

$$\widehat{H_{\text{SR}_i}} = \text{comp}(H_{\text{UP}}) \tag{15}$$

where comp() is denoted as the compact operation, which adopts 1×1 convolution. H_{SR_i} is the output HSI-SR of the ith band.

C. Bands Concatenation Module

Because the SR images of each band are generated separately, it is necessary to concatenate the images of each band into the overall SR result after an image is processed. The specific processing is as follows:

$$\widehat{H_{\text{SR}}} = \text{CON}([H_{\text{SR}_1}, H_{\text{SR}_2}, \ldots, H_{\text{SR}_b}, \ldots, H_{\text{SR}_B}]) \tag{16}$$

where $\widehat{H_{\text{SR}}}$ is the final output whole SR-HSI and CON() is denoted as the concatenate operation among all the bands. Such an operation does not add excessive spectral distortions due to the output being a single band. Because of the processing of each band, we take into account the spectral correlation between neighboring bands by the way of input HSI.

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D. Loss Function

In this article, l_1Loss() is used to calculate the mean absolute error between $\widehat{H_{\mathrm{SR}_i}}$ and H_{HR_i}, which is formulated as follows:

$$\mathrm{loss}_i(\widehat{H_{\mathrm{SR}_i}};\theta|H_{\mathrm{HR}_i}) = \frac{1}{n}\sum_{i=1}^{n}\left\|H_{\mathrm{HR}_i} - \widehat{H_{\mathrm{SR}_i}}\right\|_1 \tag{17}$$

where H_{HR_i} denotes the ith band ground truth (GT) and θ is parameter of the model. Therefore, the objective function can be described as follows:

$$\arg\min_{\theta}\mathrm{loss}_i(\widehat{H_{\mathrm{SR}_i}};\theta|H_{\mathrm{HR}_i}). \tag{18}$$

In the training process, the Adam optimizer is used to optimize the parameter θ to make the network converge well.

IV. Experiment

Extensive experiments are conducted to evaluate the performance of the proposed AD-DFGNet, for which three different datasets are exploited. The details of the training process, the parameter setting, and analysis of experiment results are introduced in this section.

A. Datasets and Evaluation Metrics

1) CAVE Dataset: The CAVE dataset (https://www.cs.columbia.edu/CAVE/databases/multispectral/) consists of 32 hyperspectral indoor scenes images, which are taken by a cooled CCD camera, and each image with the spatial resolution of 512 × 512 and the spectral resolution of 31 bands at 10 nm from 400 to 700 nm.

2) Harvard Dataset: The Harvard dataset [50] consists of 27 HSIs with more complex scenes under artificial and mixed illumination, including books, bicycles, posters, and many other hybrid scenes. Furthermore, the number of pixels is 1040 × 1392 and the spectral resolution of 31 bands at 10 nm from 420 to 720 nm.

3) Foster Dataset: The Foster dataset [51] contains 30 HSIs of outdoor natural scenes in the Minho region, Portugal, with each image collected at a spectral resolution of 10 nm for a total of 33 bands from 400 to 720 nm, and the number of pixels is 1344 × 1024.

4) Washington DC Mall Dataset: The Washington DC (WDC) Mall[1] dataset is acquired by the HYDICE sensor. There are 1280 × 307 pixels and 191 bands covering the wavelength range of 400–2400 nm, and the scene contains rich information, including rooftops, streets, grass, and so on. The WDC Mall dataset is applied to measure the performance of different algorithms in real-world scenarios.

Evaluation Metrics: Five objective evaluation metrics are used to verify the spatial and spectral reconstruction performance of the proposed algorithm: the peak-signal-to-noise ratio (PSNR), the structural similarity (SSIM), the spectral angle mapping (SAM), the correlation coefficient (CC), and the erreur relative globale adimensionnelle de synthèse (ERGAS). PSNR and SSIM are both popular evaluation metrics for evaluating image reconstruction quality. SAM focuses on assessing the spectral reconstruction distortions of the HSI, while CC and ERGAS evaluate the overall reconstruction performance of the image from a global perspective. These three evaluation metrics can be formulated as follows:

$$\mathrm{SAM}(\widehat{H_{\mathrm{SR}}}, H_{\mathrm{HR}}) = \frac{1}{\mathrm{HW}}\sum_{i=1}^{\mathrm{HW}}\arccos\left(\frac{\left\langle\widehat{H_{\mathrm{SR}_i}}, H_{\mathrm{HR}_i}\right\rangle}{||H_{\mathrm{HR}_i}||_2||\widehat{H_{\mathrm{SR}_i}}||_2}\right) \tag{19}$$

$$\mathrm{CC}(\widehat{H_{\mathrm{SR}}}, H_{\mathrm{HR}}) = \frac{1}{C}\sum_{i=1}^{C}\frac{\mathrm{cov}(\widehat{H_{\mathrm{SR}_i}}, H_{\mathrm{HR}_i})}{\sigma_{\widehat{H_{\mathrm{SR}_i}}}\sigma_{H_{\mathrm{HR}_i}}} \tag{20}$$

$$\mathrm{ERGAS}(\widehat{H_{\mathrm{SR}}}, H_{\mathrm{HR}}) = 100r\sqrt{\frac{1}{C}\sum_{i=1}^{C}\left(\frac{\mathrm{RMSE}_i}{\mathrm{mean}(\widehat{H_{\mathrm{SR}_i}})}\right)^2} \tag{21}$$

where W and H are denoted as the width and the height of HSI, respectively. In addition, C is defined as the number of HSI's bands.

As for the performance verification in real-world scenarios, GT is not accessible, and only nonreference metrics can be used to measure the performance. Three nonreference metrics, including spatial and spectral entropy quality (SSEQ) assessment [52], spatial frequency (SF) [53], and entropy, are utilized to evaluate the quality of the reconstructed HSI. The value of SSEQ usually ranges from 0 to 100 (0 for best quality and 100 for worst). For SF and entropy, larger values represent better quality.

B. Training Details

For the design of the network model, the number of DFG blocks is set as $T = 8$. In the CAVE dataset, 24, 4, and 4 images out of 32 images are used in the training set, validation set, and test set, respectively. The Harvard and Foster datasets are grouped in the same strategy, with the difference that they have three images for both the validation set and the test set due to the number of images. To facilitate the training, we divided the training sets in the three datasets into 32 × 32 LR-HSIs with a step size of 16 and performed 50 epochs with NVIDIA RTX 3070 GPU.

During the training process, $L1$ loss is used to calculate the pixel-level variance between the reconstruction result and the GT. Adam optimizer of stochastic gradient descent with $\beta_1 = 0.9$ and $\beta_2 = 0.999$ is adopted to optimize the parameters of the network, and the learning rate is initialized as $1e-4$.

C. Ablation Study

To verify the impact of the dual guidance mechanism on the performance of the network model, this section conducts ablation experiments on the CAVE dataset for two different guidance strategies. Table I shows the model performance on the PSNR and SAM evaluation metrics with and without the two modules, respectively. By comparing the metrics value, without imposing two modular constraints, it is difficult to concentrate the information of the three adjacent bands to provide sufficient a prior for the reconstruction of the middle bands, after the dual guidance operation produces a significant improvement in the performance of the network. Benefiting

[1]http://engineering.purdue.edu/ biehl/MultiSpec/hyperspectral.html

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TABLE I
ABLATION EXPERIMENTS ON FAG AND GTA MODULES UNDER CAVE DATASET IN A UPSCALING FACTOR OF 2

Components	Different combinations of FAG and GTA						
FAG	✗	✓	✗	✓(Add)	✓(Concat)	✓(Add)	✓(Concat)
GTA	✗	✗	✓	✓(Add)	✓(Concat)	✓(Concat)	✓(Add)
PSNR (↑) on CAVE(2×)	47.4207	47.7970	47.8230	47.5237	**47.9332**	47.8740	**47.9332**
SAM (↓) on CAVE(2×)	1.9737	1.8993	1.8870	1.9497	1.8831	1.8921	**1.8828**

TABLE II
ABLATION EXPERIMENTS ON GRADIENT AND TEXTURE MODULES UNDER CAVE DATASET IN A UPSCALING FACTOR OF 2

Components	Metrics	
	PSNR (↑)	SAM (↓)
grad only	47.9013	1.8903
text only	47.7291	1.8871
grad/text both	**47.9332**	**1.8828**

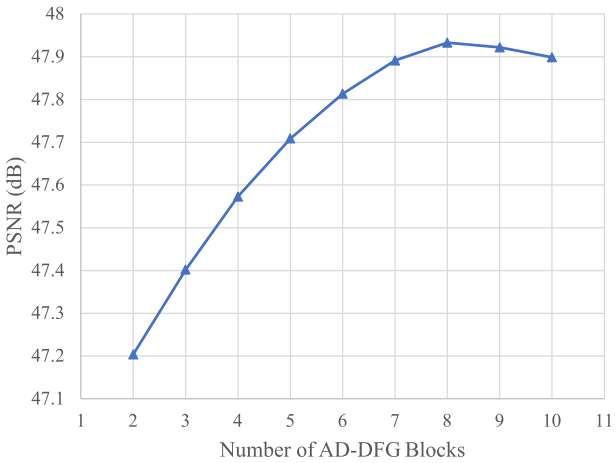


Fig. 5. Ablation experiments on number of AD-DFG blocks.

to the dual guidance block of the ith band, the network can extract features in depth and guide the reconstruction branch to focus on the gradient–textural information of the HSI. By analyzing the metrics of SAM, the results show that the FAG block and the GTA block have improved the spatial quality of PSNR, which indicates that the high-quality pixel-level spatial reconstruction has a positive effect on the spectral continuity at the same time.

Five ablation experiments are utilized to measure the performance of the proposed method in different situations. The details are described as follows.

1) FAG Block: Compared with the unguided case, the PSNR and SAM results are significantly improved when the FAG block is added alone by 0.3763 dB and 0.0744 in PSNR and SAM, respectively. The multiconvolution layers extract and aggregate the features in the individual ith band and play an effective role in the reconstruction branch. The guidance by the FAG is not sufficient to provide the best performance for the reconstructed branches. Therefore, the influence of gradient and texture information on the image SR reconstruction is considered.

2) GTA Block: The module makes full use of the gradient information and textural information of the image, and on this basis, the SR reconstruction process is precisely guided precisely guiding the reconstruction branches through the attention mechanism. When using the GTA block alone, PSNR and SAM outperform of using FAG independently by 0.026 dB and 0.0123. On the one hand, the experiment indicates that the GTA has more advantages in extracting spatial features of images, and on the other hand, it proves that GA and TA have more potential in SR tasks of HSI. The advantages of combining the FAG and the GTA in the DFG block benefit for image feature extraction in the ith band. The module guides the reconstructed branches in two separate stages, fully integrating the spatial features of this band and enhancing the spatial detail information of SR processing. From the metrics in Table I, it can be seen that the DFG block achieves the best results, which are obtained in the ablation experiments.

3) Ablation Study of Different Guidance Strategies: The two operations, named Concat and Add, are two common approaches for CNNs to fuse special features. The FAG block and the GTA block use different approaches to guide the backbone network, which are discussed in this part. As shown in Table I, with both FAG and GTA using Add operation, the performance of the network is close to that without any guidance. Conversely, in the case where Concat is used for both, the performance of the network gains significantly, as the Concat operation allows for more learnable parameters when fusing features. In addition, when alternating between Add and Concat guidance, the FAG and GTA with Concat and Add, respectively, are able to achieve the same effectiveness in PSNR as both using Concat operation and even higher in SAM by 0.0003. The guidance strategy in the proposed AD-DFGNet is an improvement in the evaluation metrics with reduced parameters.

4) Ablation Study of Gradient and Texture: In the GTA block, we measure the importance of two image structures by ablation experiments, as shown in Table II. When the gradient map is set as an all-zeros matrix, which is the case of text only, PSNR and SAM are below the optimum values by 0.0319 dB and 0.0075, respectively. On the other hand, the network performance is still limited by using grad only. Experimental evidence is that the combined effect of gradient and texture is pivotal to the enhancement of the GTA block.

5) Ablation Study of Different Numbers of Adjacent Bands: As shown in Table III, a total of five factors are selected for the ablation experiment of different numbers of adjacent bands. Bold indicates the best value, and underlining indicates the second-best value. PSNR and SAM are chosen to compare the performance of the networks in spatial reconstruction and spectral reconstruction, respectively. Experiments are conducted with two bands as a baseline, and the performance

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TABLE III
ABLATION EXPERIMENTS OF DIFFERENT NUMBERS OF ADJACENT BANDS UNDER THREE DATASETS IN A UPSCALING FACTOR OF 2

Datasets	Numbers of adjacent bands									
	2		4		6		8		10	
	PSNR (↑)	SAM (↓)	PSNR (↑)	SAM (↓)	PSNR (↑)	SAM (↓)	PSNR (↑)	SAM (↓)	PSNR (↑)	SAM (↓)
CAVE	**47.9332**	**1.8828**	47.8760	1.8893	47.8996	1.8990	47.2348	1.9013	46.9993	1.9890
Harvard	**49.5624**	2.1696	49.0324	**2.1612**	49.3260	2.1643	49.0196	2.1725	48.7855	2.2200
Foster	**51.2061**	3.6194	51.0071	**3.6161**	51.1216	3.6179	50.4725	3.7216	50.1149	3.8377

of the network changed when the number of adjacent bands increased: contrary to the theoretical results, the SAM results exactly get better with an increase in the number of adjacent bands. The reason for result above is that each band is processed with a large number of adjacent band inputs that generate redundant information, which instead reduces the efficiency of the network in extracting spectral feature. In addition, the feature extraction for spatial reconstruction is expected to focus on individual band, which means that a large number of adjacent bands is detrimental to the spatial extraction of middle band. It has been demonstrated that the number of adjacent bands is set as 2 to obtain sufficient spatial reconstruction performance while maintaining the spectral correlation of the HSI.

In addition, we use different numbers of DFG blocks in ablation experiments to explore the best performance of the network. Too few number of blocks are not enough to fully learn the features of HSI, while too many blocks make the whole network performance overfit and not optimal. When the number of blocks is set as 2–5, the performance of the model can be improved, which means that increasing the number of blocks can improve the performance of the model. Theoretically, the deeper the layers are, the better the performance of the model is. Furthermore, the dataset has an impact on optimal solution of the network. As the case of T is set as 9 and 10, the performance decreases as the network deepens. Meanwhile, it is also noted that the reconstruction performance is comparatively stable to the variation of the number from 7 to 9. In this way, to make the proposed method more flexible, the number is set as 8 universally, as shown in Fig. 5.

D. Comparison Methods

Seven classical and state-of-the-art methods in single HSI SR tasks, which are bicubic, low tensor-train rank (LTTR) [16], SRCNN [22], 3D fully convolutional neural network (3DFCNN) [23], grouped deep recursive residual network (GDRRN) [27], residual structure-texture dense network (RSTDN) [49], and spectrum and feature context super-resolution (SFCSR) [40], are compared with our proposed method. Bicubic, a traditional upsampling method using interpolation, is often used as a baseline in comparison methods. LTTR is a newer approach to machine learning that does not require the use of labels for end-to-end learning. SRCNN is a classic SR algorithm in deep learning, which contains only three convolutional layers but achieves good results. The 3DFCNN and GDRRN use 3-D convolution on top of SRCNN to maintain the spectral correlation of the HSI, further enabling mitigation of spectral distortion during spatial SR. RSTDN introduces gradient and structure information during SR, which maintains the spatial reconstruction of the HSI. SFCSR, the state-of-the-art deep learning method in the HSI-SR task, uses three adjacent bands as input in the process of spatial–spectral reconstruction, which largely facilitates the performance of HSIs in spatial reconstruction and also gives us inspiration on the input strategy.

E. CAVE Dataset

The CAVE dataset is a high-quality HSI dataset of spatial resolution. The PSNR value is the lowest compared with the other two datasets at the same SR magnification. On the contrary, the SAM value is the best, because a high-quality HSI indicates less information of each pixel and assists in reducing the interspectral error. In comparison with other methods on the CAVE dataset, the proposed AD-DFGNet shows the best performance in most cases, which is shown in Table IV. Fig. 6 shows the reconstructed image where different methods and the corresponding GT for the image flower_ms at the upscaling factor of 2. It is noted that to highlight the difference between the reconstructed image and the GT, the absolute errors for the image flower_ms at the upscaling factor of 2 are shown in Fig. 6.

All the visualization results are shown in Fig. 6. Bicubic and LTTR, as conventional methods, obtain significant errors in the petals area where there are more details in the image, and LTTR has a certain amount of grid effect. The reconstruction result of SRCNN shows smoother spatial detail reconstruction, which is a significant improvement compared with the traditional method. The 3DFCNN uses 3-D convolution to maintain the interspectral similarity of HSIs based on SRCNN and also has better spatial performance, with PSNR and SAM improving 1.8758 dB and 0.9042 over SRCNN, respectively. The parameters of GDRRN are shared, but it uses a round-robin structure to increase the computational complexity. The performance is not ideal in the palette area below the image, as seen in Fig. 6. RSTDN utilizes both the texture and structure information of the HSI with RDN as the backbone. Despite it having better results in spatial reconstruction, the depth of RDN does not show an advantage in SAM, which is higher than bicubic 0.4427. Compared with SFCSR, which is the best method for HSI-SR, the proposed AD-DFGNet has smaller reconstruction errors on the left of petal and the bottom of palette in Fig. 6. In addition, as shown in Table IV, except for CC, SAM, ERGAS, PSNR, and SSIM are 0.0521, 0.0159, and 0.1538 dB, and 0.0002 higher than SFCSR, respectively. The performance of RSTDN in ERGAS

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TABLE IV
AVERAGE MEASUREMENT METRICS FOR THE CAVE DATASET

Factor	Metrics	Bicubic	LTTR	SRCNN	3DFCNN	GDRRN	RSTDN	SFCSR	Proposed
2×	CC (↑)	0.9981	0.9978	0.9985	0.9990	0.9991	0.9990	**0.9994**	**0.9994**
	SAM (↓)	2.3344	2.3609	3.2324	2.3282	2.3061	2.7771	1.9349	**1.8828**
	ERGAS (↓)	4.6124	5.0802	4.1068	3.3844	3.1061	3.4316	2.5982	**2.5823**
	PSNR (↑)	42.5489	41.6101	43.4588	45.3346	46.0486	45.0293	47.7794	**47.9332**
	SSIM (↑)	0.9896	0.9904	0.9877	0.9931	0.9933	0.9910	0.9954	**0.9956**
3×	CC (↑)	0.9958	0.9950	0.9933	0.9962	0.9977	0.9971	0.9981	**0.9982**
	SAM (↓)	2.8973	3.0104	3.6033	2.9599	2.8974	3.5175	2.4735	**2.3954**
	ERGAS (↓)	4.6447	5.0410	4.1723	6.5675	5.1034	**3.8639**	4.6708	4.5675
	PSNR (↑)	38.8453	38.1309	39.8498	39.4563	41.7243	40.4610	42.6304	**42.8277**
	SSIM (↑)	0.9788	0.9782	0.9781	0.9803	0.9849	0.9802	0.9884	**0.9887**
4×	CC (↑)	0.9932	0.9927	0.9951	0.9958	0.9961	0.9954	0.9965	**0.9966**
	SAM (↓)	3.4404	3.4460	4.3295	3.1112	3.2989	4.2670	2.8686	**2.8251**
	ERGAS (↓)	4.4440	4.5899	3.7863	6.9444	6.6996	**3.6415**	6.3364	6.1894
	PSNR (↑)	36.7106	36.3931	38.1703	39.0706	39.4281	38.5754	40.0886	**40.2837**
	SSIM (↑)	0.9676	0.9682	0.9689	0.9773	0.9771	0.9703	0.9805	**0.9810**

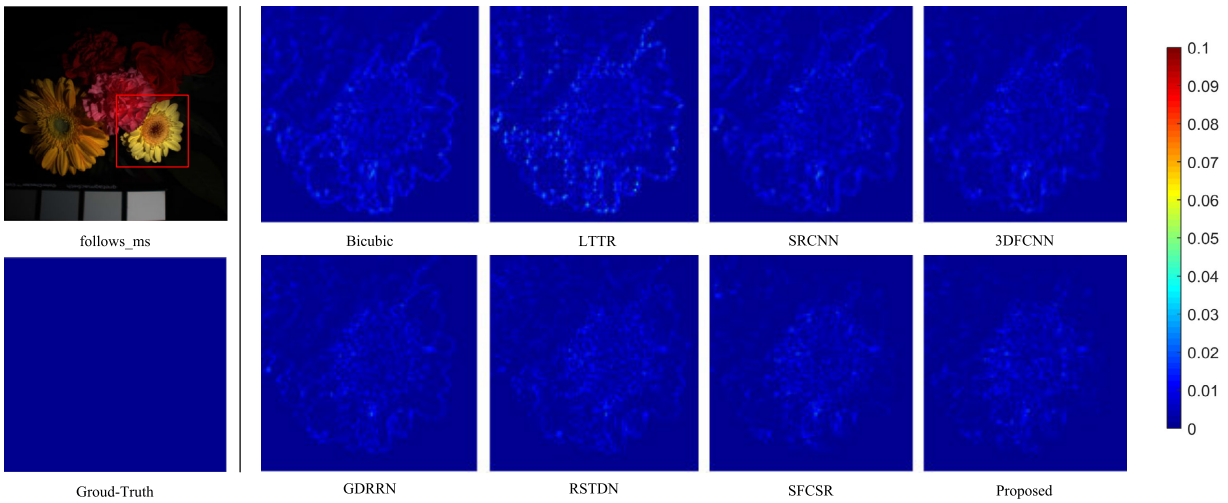


Fig. 6. Absolute error map of the reconstructed CAVE dataset at the upscaling factor of 2. The red selected area is enlarged to demonstrate, and the deeper shades of blue indicate better performance, and that the color closeness to the GT reflects the quality of the reconstruction.

TABLE V
AVERAGE MEASUREMENT METRICS FOR THE HARVARD DATASET

Factor	Metrics	Bicubic	LTTR	SRCNN	3DFCNN	GDRRN	RSTDN	SFCSR	Proposed
2×	CC (↑)	**0.9977**	0.9847	0.9799	0.9863	0.9862	0.9823	0.9872	0.9874
	SAM (↓)	3.7779	2.3326	2.7211	2.2466	2.2785	2.3620	2.1763	**2.1696**
	ERGAS (↓)	4.3342	4.0698	4.4753	3.7097	3.6963	4.1299	3.5778	**3.5243**
	PSNR (↑)	46.9660	46.3391	47.6449	48.3841	48.9707	48.8363	49.5228	**49.5624**
	SSIM (↑)	0.9955	**0.9972**	0.9953	0.9965	0.9967	0.9964	0.9971	0.9971
3×	CC (↑)	**0.9954**	0.9791	0.9770	0.9820	0.9818	0.9783	0.9830	0.9833
	SAM (↓)	4.6359	2.6330	2.8488	2.5401	2.5582	2.7198	2.4663	**2.4568**
	ERGAS (↓)	6.1140	3.2849	4.8537	4.3863	4.3385	**3.1382**	4.3172	4.1628
	PSNR (↑)	44.0943	43.6453	45.3628	45.6958	46.3474	46.0895	46.7697	**46.8673**
	SSIM (↑)	0.9919	**0.9953**	0.9930	0.9941	0.9945	0.9939	0.9950	0.9951
4×	CC (↑)	**0.9931**	0.9748	0.9740	0.9782	0.9777	0.9743	0.9789	0.9799
	SAM (↓)	5.3028	2.8127	3.0755	2.7136	2.7488	2.9822	2.6570	**2.6118**
	ERGAS (↓)	7.4828	2.7714	5.4399	4.9479	4.9285	**2.6812**	5.0449	4.6722
	PSNR (↑)	42.3147	42.0456	43.5213	43.8862	44.2393	44.2183	44.2947	**44.8824**
	SSIM (↑)	0.9884	0.9922	0.9900	0.9915	0.9917	0.9911	0.9923	**0.9928**

at the upscaling factor of 3 and 4 is the best, but the other metrics are not outstanding, and the proposed AD-DFGNet can still maintain the best effectiveness.

The visualization results of the CAVE dataset for the different four test images under the upscaling factor of 4 are shown in Fig. 7. To enhance the visual effect, specific

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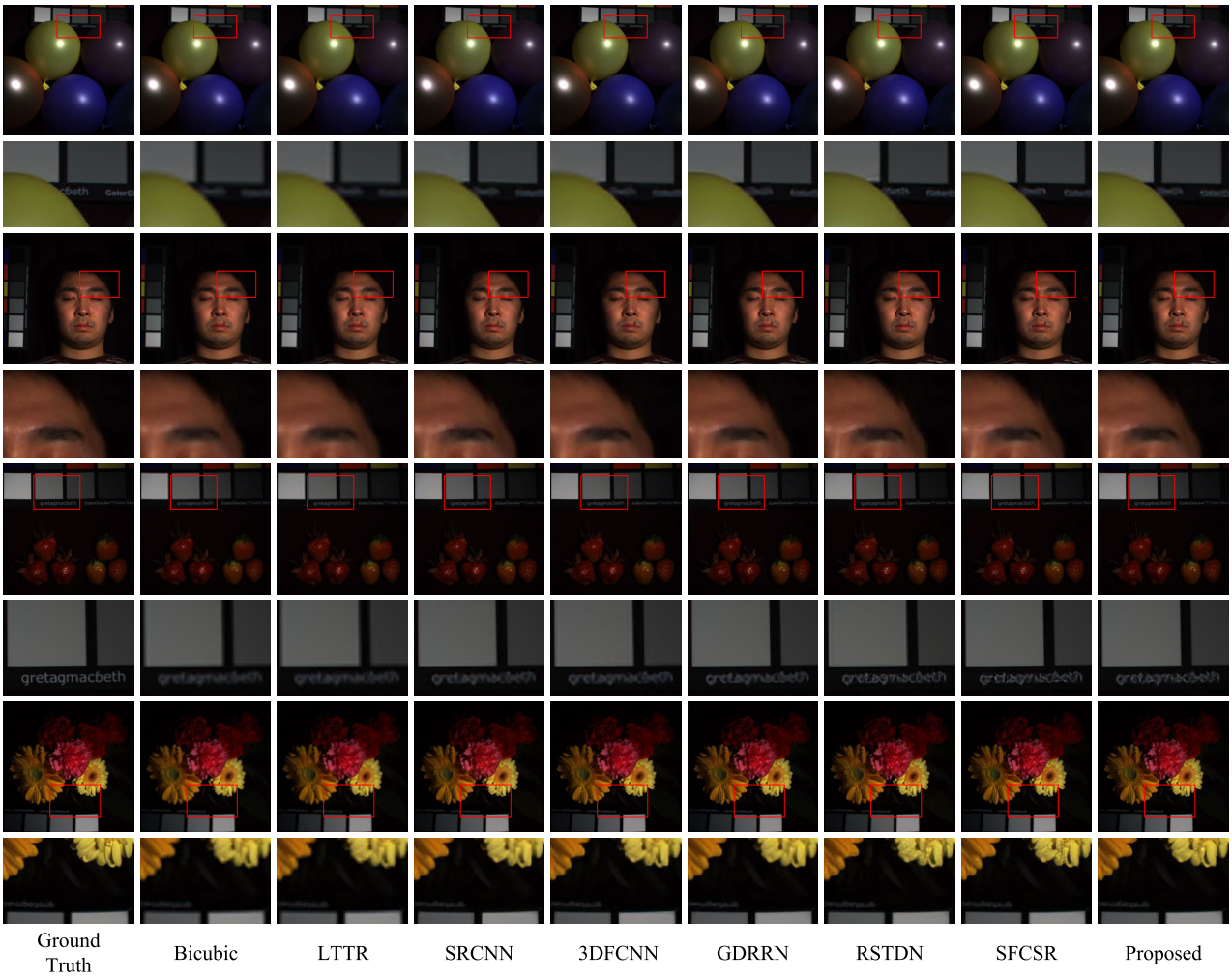


Fig. 7. Visual exhibition of the reconstructed CAVE dataset at the scaling factor of 4.

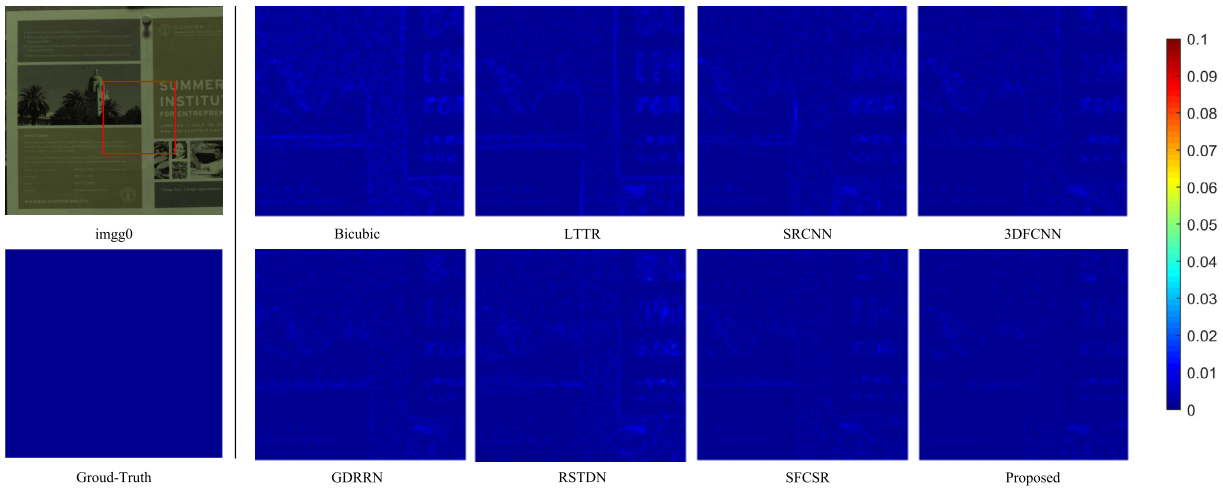


Fig. 8. Absolute error map of the reconstructed Harvard dataset at the scaling factor of 3. The red selected area is enlarged to demonstrate, and the deeper shades of blue indicate better performance, and that the color closeness to the GT reflects the quality of the reconstruction.

regions in the images are enlarged and displayed below each figure. The proposed AD-DFGNet has the best details on image edges, faces, text, and textures in the selected regions. Specifically, in the image balloons_ms and the image fake_and_real_strawberries_ms, the reconstruction of the edges and the text of the balloons reflect the SR performance of the different SR algorithms. In the image flower_ms, the proposed AD-DFGNet achieves substantial advantages

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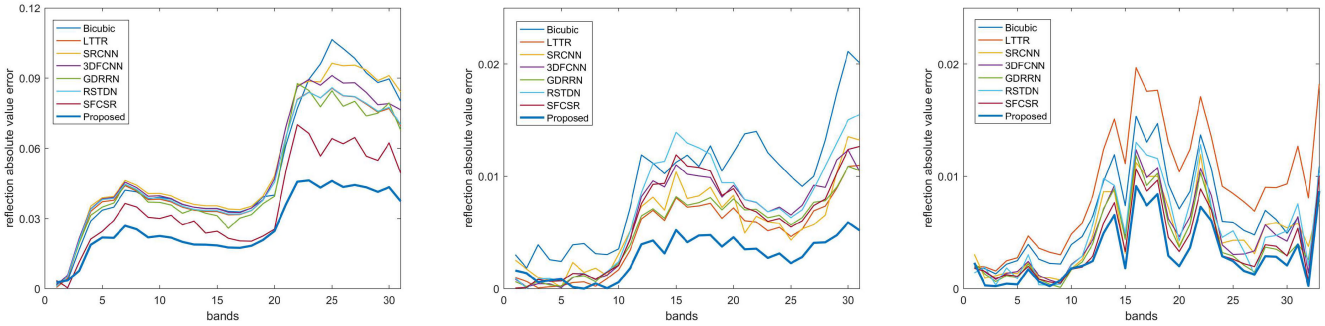


Fig. 9. Visual exhibition of the spectral curves of a randomly selected pixel in the reconstructed different dataset. (a) CAVE 2×. (b) Harvard 3×. (c) Foster 4×. The closer to the bottom horizontal coordinate indicates that the spectral reconstruction error is smaller, in contrast the larger the error is, the worse the performance is.

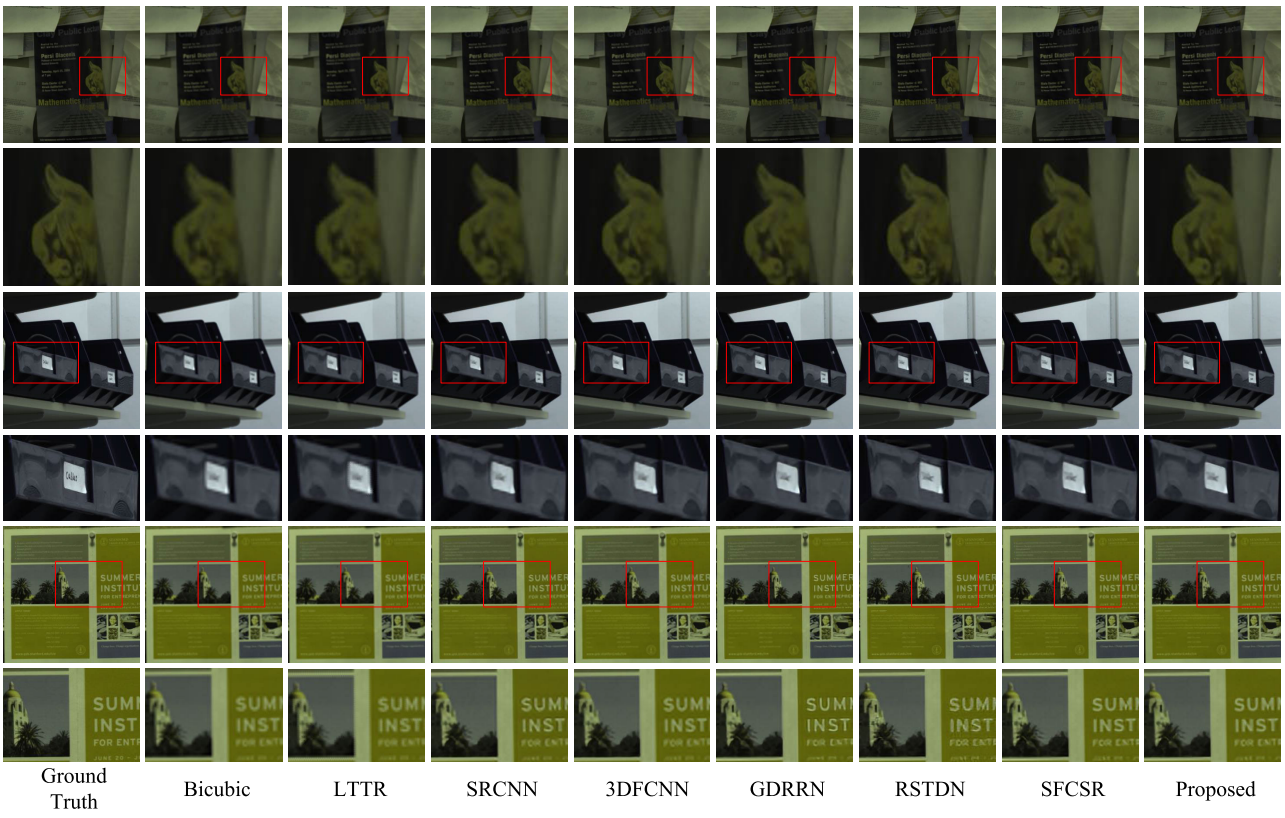


Fig. 10. Visual exhibition of the reconstructed Harvard dataset at the upscaling factor of 4.

in the reconstruction of petal edges and details. The above analysis illustrates the robustness and effectiveness of the proposed AD-DFGNet.

F. Harvard Dataset

The Harvard dataset is a real-world HSI dataset with lower spatial resolution than the CAVE dataset. Table V shows that the overall trend of the evaluation metrics is similar to the CAVE dataset.

As shown in Table V, LTTR does not perform well in PSNR results while having some advantages in SAM. Benefiting from the use of 3-D convolution, 3DFCNN and GDRRN have higher SAM values than LTTR by 0.086 and 0.0541, respectively. SFCSR as a benchmark comparison algorithm achieves better results than other comparison methods in both spatial and spectral reconstructions, especially in PSNR and SAM. The proposed AD-DFGNet achieves the best metrics in SAM and PSNR, which are higher than the second optimal value of 0.0067 and 0.0396 dB, respectively. Though the reconstruction quality of each method decreases with the increase of upscaling factor, the proposed AD-DFGNet still has a significant advantage over other methods in terms of spatial and spectral reconstruction, with the key indicators PSNR and SAM 2.5677 dB and 2.691 higher than those of bicubic, respectively. It shows that the proposed AD-DFGNet has more outstanding robustness in the high SR task.

The visualizations of the absolute error in the Harvard dataset with the upscaling factor of 3 are shown in Fig. 8. The 3DFCNN, GDRRN, RSTDN, and SFCSR are significantly

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TABLE VI
AVERAGE MEASUREMENT METRICS FOR THE FOSTER DATASET

Factor	Metrics	Bicubic	LTTR	SRCNN	3DFCNN	GDRRN	RSTDN	SFCSR	Proposed
2×	CC (↑)	0.9935	0.9905	0.9891	0.9941	0.9942	0.9943	**0.9953**	**0.9953**
	SAM (↓)	3.6802	4.3384	5.2062	3.8953	4.0951	4.1107	3.6243	**3.6194**
	ERGAS (↓)	4.8658	5.4564	6.7119	3.8172	3.8601	4.1019	3.5492	**3.5177**
	PSNR (↑)	48.2912	46.4014	45.5392	50.0264	50.5669	49.6059	50.6228	**51.2061**
	SSIM (↑)	0.9921	0.9884	0.9848	0.9941	0.9939	0.9932	**0.9949**	**0.9949**
3×	CC (↑)	0.9851	0.9806	0.9797	0.9872	0.9875	0.9855	**0.9886**	**0.9886**
	SAM (↓)	5.9396	5.4918	6.0340	5.1502	5.2750	5.7007	5.0351	**5.0069**
	ERGAS (↓)	7.5875	5.2652	11.2185	6.0862	5.9113	**4.5636**	5.6534	5.6332
	PSNR (↑)	44.0905	43.0575	42.5573	45.7922	46.3253	44.9113	46.7495	**46.8510**
	SSIM (↑)	0.9817	0.9823	0.9734	0.9852	0.9862	0.9812	0.9872	**0.9873**
4×	CC (↑)	0.9750	0.9700	0.9679	0.9785	0.9795	0.9757	0.9805	**0.9809**
	SAM (↓)	8.0020	6.2312	7.0229	5.9066	6.0706	6.6800	5.8481	**5.8230**
	ERGAS (↓)	9.9918	5.0102	15.0035	**4.0763**	7.9343	4.6486	7.6555	7.5668
	PSNR (↑)	41.3808	40.8416	39.6562	42.9764	43.4146	42.0674	43.6620	**43.9143**
	SSIM (↑)	0.9698	0.9699	0.9571	0.9748	0.9764	0.9684	0.9772	**0.9781**
	Para.(1.0e6)	/	/	**0.23**	1.15	0.37	22.3	1.23	1.84

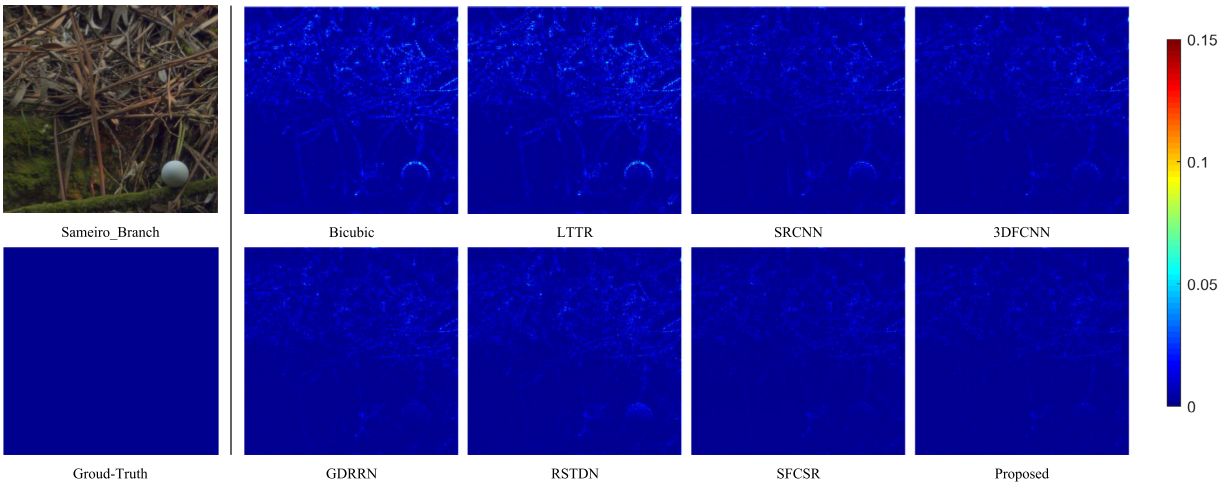


Fig. 11. Absolute error map of the reconstructed Foster dataset at the upscaling factor of 4. The deeper shades of blue indicate better performance, and that the color closeness to the GT reflects the quality of the reconstruction.

better than the above methods in spatial reconstruction details and appear sporadically rather than in a blurred manner at a large loss, especially SFCSR in detail on the relatively more stable performance. The absolute error map of the proposed AD-DFGNet is visually closer to GT, the reconstruction details of the text and images in the poster are more prominent, and the overall quality is more clear, which has better results than other comparison methods. To compare the performance of each algorithm in the spectral dimension, Fig. 9 shows the spectral reconstruction error of a randomly selected point in the image with the upscaling factor of 3, and the blue thick line indicates the proposed AD-DFGNet, which keeps the minimum error with GT overall.

Visualization results of the three HSI dataset at the upscaling factor of 4 are shown in Fig. 10. The results of almost every algorithm show different degrees of blurring at the upscaling factor of 4. Benefiting to use GTA block for image edge reconstruction, the proposed AD-DFG realizes the best results compared with the other algorithms, both in the processing of the finger parts, the paper edges, and the reconstruction of text and scene details in the poster.

G. Foster Dataset

The difference between the Foster dataset and the experimented above dataset: the Foster dataset is all complex outdoor scenes, which contain richer and more diverse information.

As shown in Table VI, the proposed AD-DFG has the best objective evaluation metrics at the upscaling factor of 2, with CC, SAM, ERGAS, PSNR, and SSIM are 0.001, 0.0049, 0.0315, and 0.5833 dB, and 0.0008 higher than the suboptimal values, respectively. In addition, the PSNR values are 0.1015 and 0.2523 dB higher than the suboptimal values at the upscaling factors of 3 and 4, respectively. The performance of the proposed AD-DFGNet does not degrade due to the increase of upscaling factors and maintains the advantage in spatial reconstruction at the same time. In addition, the results of SAM also show that the proposed AD-DFGNet has the lowest spectral loss in the reconstructed image, and the improvement of SAM for suboptimal values is 0.0282 and 0.0251 at the upscaling factors of 3 and 4, respectively. Moreover, the proposed method has a smaller number of parameters and better performance compared with RSTDN. For the Foster dataset in Table VI, when compared with

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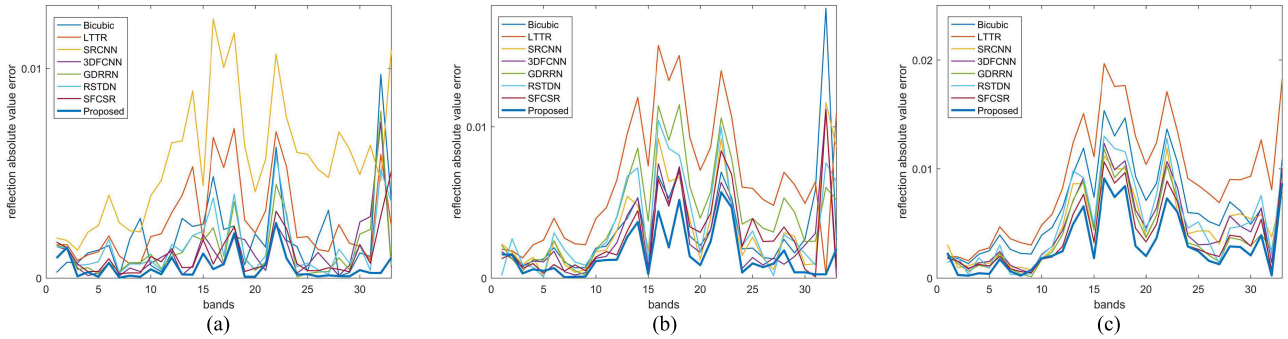


Fig. 12. Visual exhibition of the spectral curves of a randomly selected pixel in the reconstructed Foster dataset. (a) Foster 2×. (b) Foster 3×. (c) Foster 4×. The closer to the bottom horizontal coordinate indicates that the spectral reconstruction error is smaller, in contrast the larger the error is, the worse the performance is.

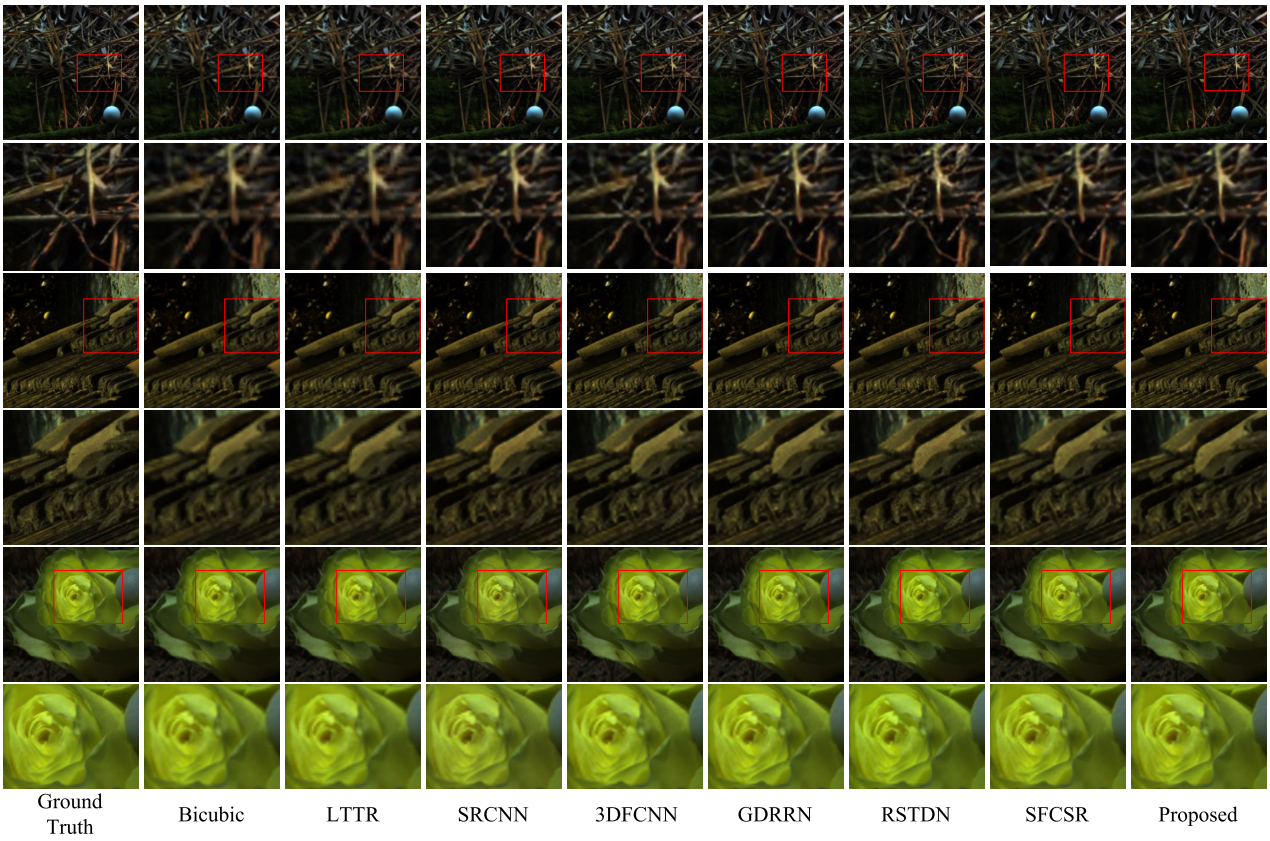


Fig. 13. Visual exhibition of the reconstructed Foster dataset at the scaling factor of 4.

the SFCSR whose parameter number is 1.23 million, the proposed method achieves a 0.2523-dB improvement with 1.84 million parameters when the scaling factor is 4. As to the scaling factors 2 and 3, the proposed method still achieves a 0.5833- and 0.1015-dB improvement in PSNR. In addition, the proposed method achieves much more effective performance over the other competitors.

The results of the spatial absolute error visualization of the Foster dataset at the upscaling factor of 4 are shown in Fig. 11. Bicubic and LTTR cannot reconstruct the high-frequency details completely, resulting in significant errors in the visualization effect. The results of SRCNN, 3DFCNN, and GDRRN are similar in the Bamboo_Leaf part, while there is a significant difference in the processing of the low-right Ball. The RSTDN utilizes the gradient and texture information of the image for deep convolution, and the loss of the Bamboo_Leaf part is smaller, but the shortcoming is that it does not continue to process the Ball position more finely. SFCSR and the proposed AD-DFGNet maintain excellent results for the overall absorb of the HSI. Compared with RSTDN and SFCSR, the proposed AD-DFGNet achieves less loss in the left and right parts of the absolute error image than SFCSR, because the gradient and structure in attention-driven processing are more advantageous in high frequency and rich detail HSIs.

Fig. 12 shows the spectral reconstruction performance of the Foster dataset at the upscaling factors of 2–4. The proposed AD-DFGNet obtains a stable performance and always

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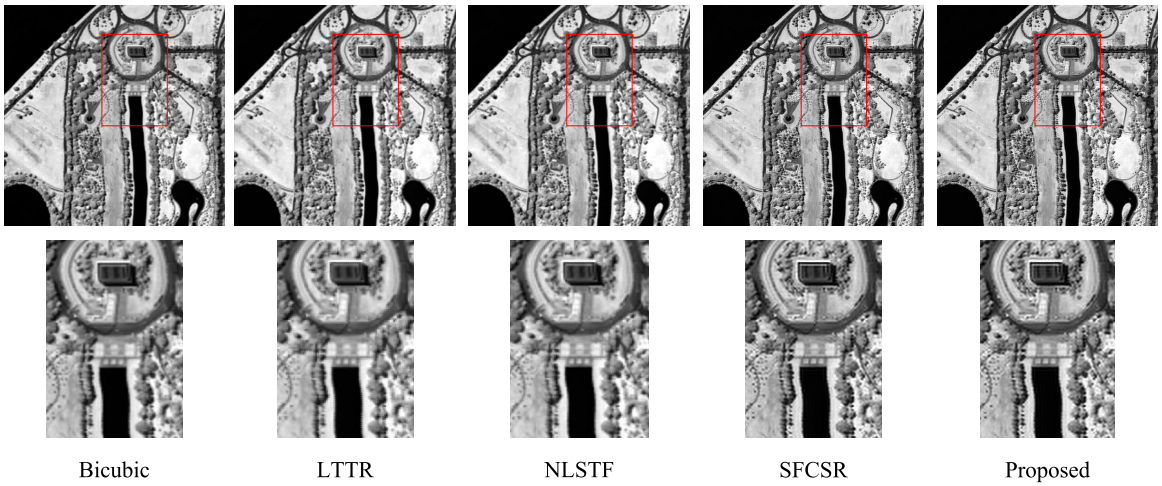


Fig. 14. Visual difference between the WDC Mall super-resolved by different methods at the upscaling factor of 2.

TABLE VII
NONREFERENCE EVALUATION METRICS FOR THE WDC MALL DATASET

Metrics	Bicubic	LTTR	NLSTF	SFCSR	Proposed
Entropy (↑)	7.4410	7.4766	7.4693	7.5605	**7.5953**
SF (↑)	22.9386	24.4813	23.3708	30.0427	**30.1337**
SSEQ (↓)	40.8356	38.9978	39.0752	33.5969	**30.7771**

maintains a relatively lower spectral reconstruction error. By analyzing the data at the upscaling factors of 3 and 4, the proposed AD-DFGNet achieves a smaller increase of SAM in multiplicity than other comparing algorithms.

Fig. 13 illustrates the visual results of the test set. Due to the dual guidance that plays the role of depth aggregation in the spatial reconstruction of the HSI, the proposed method has the best edge retention performance. It proves GTA block plays an excellent usage in the HSI edge detail and overall structure. Specifically, in the image Sameiro_Branch, the intricate branches of the tree are enlarged. The reconstruction quality of the deep learning methods is significantly improved, benefitting from end-to-end learn deep features between the LR-HSI and the GT. However, the reconstruction results may also vary significantly between different network architectures and design strategies. For instance, in the Yellow_Rose, the reconstruction quality of GDRRN, RSTDN, SFCSR, and the proposed method is clearer in the central region of the petals, especially the proposed method, because of the use of FAG block and GTA block for the extraction of spatial features in the network, such that the best spatial–spectral reconstruction is achieved.

Experimental results of three datasets on both subjective visual effects and objective evaluation metrics verified the effectiveness of the proposed AD-DFGNet for both indoor HSIs and the outdoor HSIs with poorer spatial information. Especially, for the reconstruction of complex edge details, the proposed AD-DFG has more significant superiority.

H. WDC Mall Dataset

To consider the importance and practical applications of reconstructing edge information in real-world scenarios, relevant experiments in the WDC Mall dataset are experimented to verify the effectiveness of the proposed method. As shown in Table VII, SFCSR is chosen as the deep learning method, because it has the advantage of not requiring retraining on a different dataset. Meanwhile, we have also selected the existing typical and effective traditional methods, which are bicubic, LTTR, and NLSTF. Distinguishing from RGB images, the number of bands in HSIs acquired by different devices ranges from tens to thousands. However, differences in the number of output channels make it difficult to perform convenient generalizability validation on different datasets. The proposed AD-DFGNet improves on the problems mentioned above and permits it effortless testing on different datasets. It is demonstrated experimentally that the proposed method still exhibits the optimal generalization and robustness even when the well-trained model is used for testing in real-world scenarios. Especially, in SSEQ, the proposed method is 2.8198 higher than the second best SFCSR.

As shown in Fig. 14, we magnify the red boxed area in the reconstructed HSIs. It can be clearly seen that the details on the edge of the black water are optimally shaped by the proposed method. Furthermore, in the woods and rood position on the left, both SFCSR and the proposed method demonstrate superior reconstruction capability. However, the reconstructed results of the roads reflect the ability of our method to recover much more distinct edges. Benefiting from the excellent gradient guidance provided by the GTA module, the proposed AD-DFGNet can still obtain the sharpest edge information for the real-world scenarios. Also, it is sufficient to show that the method is effective in practical applications.

V. CONCLUSION AND FUTURE WORK

In this article, we focus on using the deep feature-guided approach to enhance the spatial resolution of HSI and maintain

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the spectral information. Specifically, a novel method named AD-DFGNet is proposed, which contains two feature guidance steps. The first guidance extracts deep spatial features of the reconstructed bands. The second guidance utilizes an attention-driven gradient–texture block to enhance the edge details and overall structural quality of the reconstructed HSIs. Extensive experiments and data analysis on selected public three HSI datasets with different scaling factors have proved the effectiveness of the proposed method.

Although the proposed method can be applied to all HSIs without changing the number of the output band, the performance of generalization is still limited. In the future, we will aim to employ transfer learning strategies, such as domain adaptive, to reduce the domain gap between different datasets. On the other hand, it is effective that the gradient–texture guides the reconstruction process, and the subsequent work will be performed in the direction mentioned above for feature consistency extraction.

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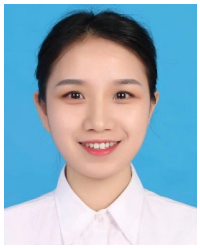
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